A structured tool to improve clinical outcomes of Type 2 diabetes mellitus patients: A randomised controlled trial

Short running title: Diabetes tool trial

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Abstract

Background: A review of pharmacist diabetes intervention studies revealed lack of structured process in providing diabetes care which consequently produced varied results from increased to minimal improvements. This study aimed to determine the effectiveness of a structured clinical guidelines tool, the Simpler™ tool, in the delivery of diabetes care. The primary outcome was significant improvement in HbA1c (glycated haemoglobin). Secondary outcomes were improved lipid profiles and blood pressure (BP).

Method: A 6-month, parallel, multi-centre, two arms, randomised controlled trial involving 14 pharmacists at seven primary care clinics was conducted in Johor, Malaysia. Pharmacists without prior specialised diabetes training were trained to use the tool. Patients were randomised within each centre to: 1) Simpler™ care (SC), receiving care from pharmacists who applied the tool (n=55); 2) Usual care (UC), receiving usual care and dispensing services (n=69).

Results: SC reduced HbA1c significantly by 1.59% (95%CI: -2.2, -0.9) compared to 0.25% (95%CI: -0.62, 0.11), (P=<0.001) in UC. In addition, SC patients had significantly improved systolic BP: (-6.28 mmHg (95%CI: -10.5, 2.0), p=0.005). The proportion of patients who reached the Malaysian guideline treatment goals were significantly more in the SC arm

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1111/jdb.12799

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(14.3% vs 1.5% for HbA1c, *p*=0.020; 80% vs 42% for systolic BP, *p*=0.001; 60.5% vs 40.4% for LDL cholesterol, *p*=0.046).

Conclusions: Use of the Simpler™ tool facilitated delivery of comprehensive evidence-based diabetes management and significantly improved clinical outcomes. The Simpler™ tool supported pharmacists in providing enhanced structured diabetes care.

Keywords: Diabetes, intervention, tool, randomised controlled trial, primary care

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Highlights of the study

This study has shown the benefits of pharmacists utilising the Simpler™ tool to make consistent, individualised, evidence-based interventions in the clinical management of patients with T2DM. Qualified pharmacists in the community or hospital settings with or without prior formal diabetes training can be trained to use the tool to deliver quality diabetes care. The findings may be generalised to studies which recruit participants with similar inclusion and exclusion criteria and with direct access to a prescriber.

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Introduction

The reported prevalence of diabetes among Malaysian adults (≥ 18 years) in 2015 was 17.5%[1]. Of the 1688 diabetes patients studied in the 2013 Malaysian Diabcare study, 15.7% had angina and 18.4% had suffered from a myocardial infarction.[1] The prevalence of neuropathy was 41% and cataracts 31.5%.[2] These findings highlight the need for healthcare professionals to provide quality care in order to minimise diabetes related complications.

The management of Type 2 diabetes mellitus (T2DM) requires a multifaceted approach. Large T2DM randomised controlled trials (RCTs) such as the United Kingdom (UK) prospective diabetes study (UKPDS) and action to control cardiovascular risk in diabetes (ACCORD) trials observed the need to achieve blood pressure and cholesterol targets in addition to glycaemia targets.[3,4] Diabetes guidelines from Australia, Malaysia, the UK and the United States of America (USA) outline an additional four diabetes factors aimed to reduce diabetes related complications.[5-8] Those are medication, lifestyle, cardiovascular risk management and patient education. Nonetheless, few studies have documented the effect of all seven diabetes factors in the management of T2DM patients.[9] As a result, diabetes interventions led by multidisciplinary teams comprised of nurses, dieticians, psychologists and physiotherapists indicated improvements in certain of the seven factors but not others.[10-16] A review of pharmacist diabetes intervention studies revealed lack of structured process in providing diabetes care which consequently produced varied results from increased to minimal improvements.[9,17]

Several studies have revealed that healthcare professionals utilise tools in making treatment choices and in providing patient counselling and education.[18-21] A tool used by physicians and other healthcare professionals for record keeping is SOAP, an acronymn for subjective,

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objective, assessment and plan where each letter of the word represents a section that is included in the case notes.[22] A previous study reported the effectiveness of a checklist consisting of 88 questions for pharmacists to use when conducting diabetes management reviews.[23] However, there is a lack of published evidence of pharmacists using this long checklist in practice. A simplified tool to manage T2DM patients with complications, which prompts pharmacists about their management, has not been documented.

Objectives of the study

The primary aim of this study was to investigate the impact of using a diabetes intervention tool that addresses the seven required diabetes factors to reduce HbA1c in T2DM patients by ≥1%. The secondary aim was to achieve significant improvements in low density lipoprotein cholesterol and systolic blood pressure (BP). This study tested the hypothesis that providing a structured intervention tool would result in significantly improved clinical outcomes in the intervention arm over the usual care arm. Although all seven diabetes factors were addressed in this RCT, this paper however will only focus on glycaemia, BP and cholesterol interventions.

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Methods

Study design and locations

The study design was a two-armed prospective, parallel group, multi-centred RCT. It was conducted between June 2016 and February 2017 in the State of Johor, Malaysia. The participating sites were seven government funded primary healthcare practices. These practices, known as health clinics (*Klinik Kesihatan, KK*), provide comprehensive medical care for ambulatory patients.[24] The research method followed the recommendations of the CONSORT statement for randomised trials[25]. A summary of the overall study process is presented in Figure 1. The Curtin University Human Research Ethics Committee (HREC) approved this study (HR214/2015) as did the Malaysian Medical Research and Ethics Committee. The trial was registered with the National Medical Research Register of Malaysia and can be accessed at www.nmrr.gov.my.

Insert Figure 1 here

Multifaceted diabetes intervention tool

A multifaceted diabetes intervention tool, referred to as the Simpler™ tool, was the basis of the study. It presented seven diabetes factors and 32 corresponding indicators for the management of T2DM. Each letter of the word "Simpler" represented a primary diabetes factor namely: **S**tatin/cholesterol; **I**nsulin/glycaemic; **M**edication; blood **P**ressure; **L**ifestyle; **E**ducation and cardiovascular **R**isk prevention strategies. These indicators were sourced

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from diabetes guidelines from Australia, Malaysia, the UK and the USA. [5-8] The content, format and layout of the tool was validated by 12 diabetes experts from Australia and Malaysia using the Delphi method.[26,27] Subsequently, a Simpler™ training package was developed and pilot tested by 12 full time community based pharmacists in Australia and Malaysia trained to use the tool in their usual practice setting.[28] The tool is presented in Table 1.

Insert Table 1 here

Recruitment of pharmacists and training

The eligibility criteria for pharmacists were having less than three years of clinical experience in diabetes management and not undergone the credentialing process to become a credentialed diabetes pharmacist by the Pharmaceutical Services Division, Ministry of Health, Malaysia. Two hour online Simpler™ training program[28] commenced once the pharmacists signed consent forms to participate in the research. This was to ensure skills uniformity among the pharmacists as suggested in previous literature.[29] All participants were briefed by the primary researcher on the protocol before commencement of the RCT.

Recruitment of patients

A power analysis using the DSS research sample size calculator[30] indicated that for the primary objective at least 52 patients were needed to show a significant mean HbA1c reduction of 1.33% between baseline and six months in the intervention group compared to

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0.4% in the control group. The standard deviation was 1.9. These parameters were selected based on findings from two previous studies conducted in the states of Johor and Selangor, Malaysia that showed a HbA1c reduction of 1.33% at the end of study periods among patients receiving diabetes medication management services from pharmacists and a HbA1c reduction of 0.4% among patients who received usual care.[31,32] A sample of 60 patients for each arm (intervention and control) was determined based upon a drop-out rate of 15%.

The inclusion criteria for patients were: T2DM diagnosis, aged > 21 years, on multiple medications including for other chronic conditions and/or have other diseases in addition to diabetes, HbA1c > 8% or fasting blood sugar > 7.0 mmol/L or two hours post prandial sugar level > 8.5 mmol/L. The exclusion criteria were: if patients had been diagnosed with dementia or severe psychotic illness, on haemodialysis treatment, pregnant or lactating, intended to become pregnant within the next six months, diagnosed with cancer, or were involved in other research at the time of recruitment.

Patients were randomised to receive the intervention, namely Simpler™ Care (SC), or be in the control and receive usual care (UC), using overall equal randomisation (1:1) at each of the seven selected sites. The randomisation numbers were predetermined using an online random number generator based on a one block randomised block design.[33] Participating pharmacists were asked to recruit 24 patients at their respective clinics during patients' scheduled visits to the doctor. Recruitment processes were conducted either through proactively targeting patients who fulfilled the inclusion criteria or through referral by clinics' diabetes nurses. The patients consulted with the Simpler™ trained pharmacist prior to visiting the doctor. Pharmacists provided to potential participants the information sheet and a brief explanation regarding the research. Upon receiving written consent,

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pharmacists opened the envelopes in ascending order and depending on the randomisation code allocated patients to either the intervention or control arms.

Study procedure

The two arms consisted of the intervention group patients (pharmacists using the Simpler™ tool and hence provide Simpler™ care, SC) or the control group patients (pharmacists providing usual care, UC). SC patients were seen at the beginning of the study and were given appointments to meet with the intervention pharmacist five more times throughout the six month period.

UC patients were also seen at the beginning of the study and were given an appointment to be seen once more at six months for the purpose of data collection. Apart from this, UC patients did not interact with the intervention pharmacists.

The study design in terms of frequency of patients' visits in both arms was based on previous diabetes intervention studies.[34-38] Differences in clinical outcomes were quantitatively measured pre-and post-intervention and between arms. Pharmacists performed blood pressure (BP) measurements using OMRON HEM/120 arm blood pressure monitors provided as part of participation in the study. Patients' blood glucose were measured using Abbott Freestyle Optium meters provided at the respective clinics. In addition to government funded and managed laboratories, HbA1c measurements were performed by a registered private laboratory.

UC patients received the usual customary care. This care included routine medical care of diabetes patients at primary health clinics consisting of doctor and nurse visits that ranged from every three to five months. In addition, all patients received pharmacy care which included monthly medication refills and counselling.

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SC patients received the usual customary care, similar to patients in the UC arm. In addition, they had a face-to-face interview of at least 20-30 minutes with the intervention pharmacist. During this time pharmacists utilised information gathered from patients and medical records to make assessments. Subsequently, by using the Simpler™ tool, pharmacists documented their interventions and individual goals for patients in the forms (Simpler™ pro forma) provided which were then monitored at each subsequent visit. Pharmacists' interventions included addressing medication related concerns and making recommendations to the prescribers. They also provided patient education based on the Simpler™ tool indicators. All seven goals were addressed at each visit. However, pharmacists were flexible regarding patient education on lifestyle factors and used their judgement in terms of addressing these issues throughout the six-months period. The Simpler™ pro forma was attached to patients' medical records and doctors were encouraged to also record their interventions using the Simpler™ pro forma. Pharmacists subsequently captured an image of each completed Simpler™ pro forma and forwarded it to the principal researcher monthly. In addition, patients in both arms completed the WHOQOL-BREF questionnaire at the first and final visits of the study (final visit). Pharmacists telephoned patients the day before appointments to remind them of their visits with the pharmacist. During patient follow up when the patient had no doctor's appointment, pharmacists informed the doctor on any therapy change required. Subsequently, all interventions were documented.

Data analysis

Data analysis was performed on primary and secondary outcomes variables using the IBM Statistical Package for Social Sciences software version 22.[39] Outcomes from the

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intervention and control arms were compared at baseline using independent t-tests for parametric data and the Mann-Whitney U-test for non-parametric data. A Pearson chi-square test was used for categorical parameters and subgroup analysis. Paired t-test was used to detect significant difference at baseline and at 6-months. Occasionally one-way analysis of variance (Anova) was used to determine significant difference between three or more categories. During the study period, alpha level of ≤ 0.05 was used for all statistical tests to denote significance level.

Results

A total of 154 T2DM patients (77 SC: 77 UC) attending the KKs were recruited into the study. Any patients not attending the final visit were designated as non-completers and therefore not included in the analysis. There were 89.6% (69/77) UC patients who completed the study. Reasons for non-completion included patient absent on follow up visits despite reminders and telephone calls, patients did not have outcome measures available or had passed away. In the SC arm, 79.2% (61/77) patients completed the study duration. The average number of visits to the pharmacist was five (SD=1.43). The number of visits ranged from two to six visits for each patient. However only 71.4% (55/77) patients attended a minimum of three visits which was the minimum number to be included in data analysis. A required minimum of three visits was agreed by the research team as this was greater than the number of control patients' visits and enabled one pharmacist reinforcement visit over the trial period. The main reason for patients completing less than the agreed six visits was not being able to meet at predetermined dates. In other instances, pharmacists were not available to meet patients who visited the pharmacy outside the predetermined dates due

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to dispensing workloads. Figure 2 is a CONSORT presentation of patients' recruitment, completion and reasons for non-completion.

Insert Figure 2 here

The baseline characteristics of patients in the SC and UC arms are shown in Table 2. A comparison of SC and UC with respect to demographic, family history, types and number of comorbidities and current employment showed no significant differences. There was a significant difference between patients' overall highest education level between the SC and UC arms. Nevertheless, the percentage of patients who were educated and those who were not did not differ among SC and UC arms (p=1.000).

Insert Table 2 here

Clinical outcomes

Table 3 presents the results of a range of clinical parameters and their individual analysis using independent t-tests in the SC and UC arms at baseline and at 6-months intervention. Results show significant improvement for SC over UC: HbA1c, -1.59% vs -0.25%, (p=<0.001); and systolic BP, -6.28 mmHg vs 0.26 mm Hg (p=0.028) at 6 months. The effect size for HbA1c differences in this research is 0.702 for Cohen's d and 0.722 for Hedges'g. In addition, the SC arm showed significant reductions in HbA1c, systolic BP, LDL and HDL at six months compared to baseline. However, in the UC arm, only the HDL measurement showed significant improvement at 6 months compared to baseline.

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There were no significant changes in the number of insulin, oral hypoglycaemic, antihypertensives and lipid lowering (statin) medications prescribed at six months of the study between the arms. However, a higher proportion of SC patients vs UC were on statins, 47/55 (85.5%) vs 52/69 (75.4%) and metformin, 40/55 (72.7%) vs 44/69 (63.8%) at six months. In contrast, more UC patients were prescribed insulin 64/69 (92.8%) over SC 47/55 (85.5%) at six months. Similarly, in accordance with the Malaysian diabetes guidelines,[6] significantly more patients in the SC arm achieved the treatment goals (refer to Appendix 1) for HbA1c (p=0.020), systolic BP (p=0.001) and LDL cholesterol (p=0.046) compared to the UC arm as shown in Table 4.

Insert Table 4 here.

An overall majority of SC patients (57.1%) achieved a mean reduction of at least 1% HbA1c compared to 28.6% in the UC arm. This difference was statistically significant (p=0.002). Patients with a mean baseline of more than 9% (p=0.008), patients of Melayu ethnicity (p=0.003) and female patients (p=0.006) had significantly improved HbA1c compared to patients in the UC arm. Details of these data are presented in Table 5.

Insert Table 5 here

Discussion

Pharmacists' interventions in the management of patients with T2DM, using the Simpler ™ tool, showed positive results as demonstrated by a significant decrease in the HbA1c and systolic BP values compared to UC patients. In addition, the Simpler™ intervention also improved other clinical outcomes such as LDL and HDL levels at six months compared to

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baseline. The magnitude of HbA1c improvements pre and post intervention measured by the effect size using Cohen's d and Hedges'g between SC and UC patients were comparable to RCTs performed among hospital outpatients[40,41] and at a primary health care clinic [42] in Malaysia.

Statin/Cholesterol control

Several published RCTs and retrospective studies conducted by pharmacists in outpatient hospital settings in Malaysia [36,43] and Hong Kong [44] have shown similar improvements in LDL to our study. Their baseline values ranged from 2.11 to 3.22 mmol/L similar to this study. In addition, the proportion of patients in the SC arm who achieved the LDL and HDL target set by Malaysian diabetes guidelines were significantly more than the patients in the UC arm. Both improvements in LDL and HDL values in the SC group are possibly associated to increased patient adherence to current statin medications and increased numbers of patients being prescribed a statin as recommended by pharmacists.

Glycaemia control

The overall improvement of HbA1c in this study was consistent with the findings from RCTs conducted in community and hospital settings in Malaysia[40,41,45] and other parts of the world.[9,17] The UKPDS trial found a mean reduction of 1% for HbA1c was associated with a 14% reduction of risk for myocardial infarction, 37% for microvascular complications and 21% for deaths related to diabetes.[46] These results from the UKPDS trial[46] could be used to

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indicate the effectiveness of the Simpler™ intervention tool as this method of using HbA1c to measure the effectiveness of an intervention has been used in previous studies.[9,17,29] In our study, pharmacists utilising the Simpler™ tool reduced HbA1c by more than 1% on average. The proportion of patients who achieved the desired mean HbA1c was significantly more in the SC arm compared to the UC arm. In addition to HbA1c values, comparison of FBG readings (baseline versus six months) showed significant improvement for the SC arm, namely a reduction of 3.76 mmol/L ($p<0.001$). In comparison, FBG readings reduced by only 0.63 mmol/L ($p=0.524$) in the UC arm. These reductions were significantly different between arms ($p<0.002$).

A significantly increased number of patients with baseline HbA1c > 9% achieved reduction of 1% for HbA1c in the intervention group compared to patients with baseline values of <9%. This finding is compatible with a systematic review on diabetes interventions conducted internationally where greater reduction in HbA1c was found with higher baseline levels.[17] It is notable that the baseline HbA1c for most patients in this study (71.0%) were >9%, which demonstrates the need for focused patient interventions.

Another interesting outcome was that significantly more patients in the intervention group from *Melayu* ethnicity and female gender achieved the mean reduction of 1% for HbA1c. Even though this study was not initially statistically powered to account for differences among different ethnicities and gender groups, the intervention arm managed to reduce diabetes related complications in this section of the population which comprised the majority of the patients recruited.

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In this study, systolic but not diastolic BP was significantly improved in the intervention group at six months compared to baseline and between groups. This is comparable with a recent RCT conducted in a hospital setting in Malaysia [36] but not in other Malaysian studies which included hospital and community settings.[40-42,47,48] The BP improvement is also consistent with a RCTs conducted among multiple primary health care centres in Brazil, Canada and the UK, [38,49,50]. This significant decrease in systolic BP is associated with a reduction in stroke or myocardial infarction risk by half and was equally as important as glycaemia control in the UKPDS trial.[4]

Numerous trials on the association of high salt intake and hypertension have been identified in a systematic review.[51] Previous studies have documented that obese and female patients tend to have increased salt intake and consequently higher blood pressure compared to patients with low salt intake.[52]The average BMI values for both intervention and control patients in this study at baseline were 29kg/m^2 which is classified as obese (BMI≥23kg/m^2).[6] Intervention pharmacists actively educated patients on reduction of salt intake as recommended by the 2015 Malaysian diabetes guidelines.[6] This could have contributed to the higher proportion of SC patients achieving the BP systolic target compared to UC. In addition, findings from the MRFIT trial showed that greater CVD risk with each increment of systolic BP and individuals with increased systolic but not diastolic were at a greater risk of stroke[53] and mortality[54].

In this study, the research team agreed to include patients who attended a follow up meeting with the pharmacist at least three times during the 6-month research period for

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data analysis. This is consistent with previous studies which found that patients showed a significant difference in their clinical outcomes at three visits of follow up.[37,40] Of interest, the HbA1c reduction was 0.79% (95%CI:-1.75%, 0.16%)(n=18) in patients who completed three to five visits during the trial period. In patients who completed the six visits, the HbA1c reduction was 2.05% (95%CI: -2.92%, -1.19%) (n=31). The HbA1c value of patients in the latter cohort showed increased glycaemic improvement than hospital outpatients who completed eight visits in a similar RCT conducted in Malaysia (reduction of 0.90% HbA1c, n=39), baseline HbA1c values (10.11%, SD=0.26).[36] Undoubtedly, the Simpler™ intervention arm managed to reduce more than 1% HbA1c in fewer visits. However, future research is warranted to establish the frequency of visits needed to reduce HbA1c value by 1% in order to determine the cost effectiveness of the Simpler™ tool.

Limitations

Not all HbA1c and lipid results were available for each patient due to laboratory tests not scheduled at the exact time before the trial commenced or at completion. Hence the values obtained, ranged from the previous month to previous four months. However, every effort was taken to obtain the most recent results before the trial commenced and on completion for both groups.

Secondly, as both intervention and control patients were recruited from the same primary care health centre, contamination in the control group could occur. This is because it is possible that the same doctor treating intervention patients would be treating the control patients and therefore may apply the pharmacist's recommendations for the intervention

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group to the control group throughout the trial period. Nevertheless, the effect of any contamination did not mask the improvements seen in the SC group.

Thirdly, The Hawthorn effect may be neutralised as patients in both arms were required to visit the pharmacy every month. However, patients in the intervention arm had longer contact time with the pharmacist than the control patients. This could have contributed to patient's improvement as was concluded in previous studies.[55,56]

Conclusions

This study has shown the benefits of pharmacists utilising the Simpler™ tool to make consistent, individualised, evidence-based interventions in the clinical management of patients with T2DM. Most notably, in improving glycaemia and systolic BP outcomes. The findings may be generalised to studies which recruit participants with similar inclusion and exclusion criteria and with direct access to a prescriber. Qualified pharmacists in the community or hospital settings with or without prior formal diabetes training can be trained to use the tool to deliver quality diabetes care.

Acknowledgements

No funding was received for this research article. The authors would like to thank the Director General of Health Malaysia for his permission to publish this article. The authors would like to acknowledge the effort, dedication and contribution of all 14 participating pharmacists; Norain Mohd.Puat, Christina Chan Yuin Theng, Nur Elyana Yazmin Mohd Redhuan, Lee Kok Pun, Nurul Huda Yasim, Ng Chen Kun, Nurul Azwa Ishak, Sharon Augustine Amburose, Nur Salwa Abdul Halim, Tan Hui Yng, Tey Sin Wei, Tan Cha Sin, Foong Yoke Kuan and Mohd Azmer Lias. The authors would also like to thank Mrs.Rosidah Md Din for her support in this research.

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Disclosures

None declared.

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Figure legends

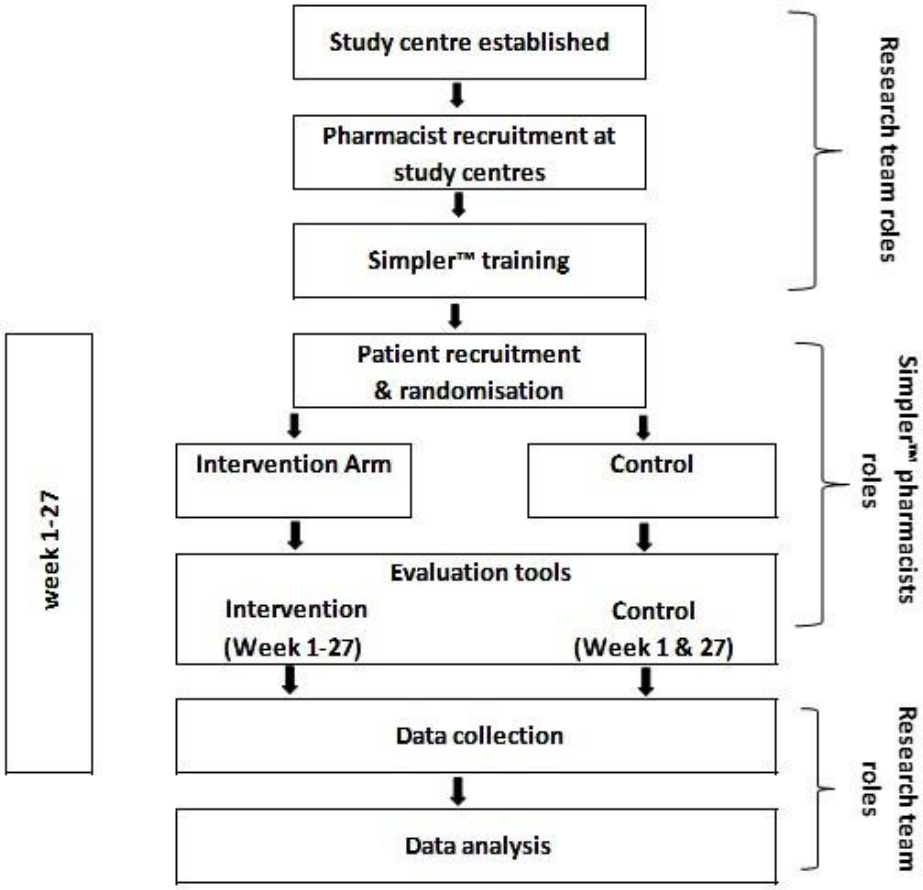

Figure 1: Flow diagram of randomised controlled trial evaluates pharmacists' interventions using the Simpler™ tool

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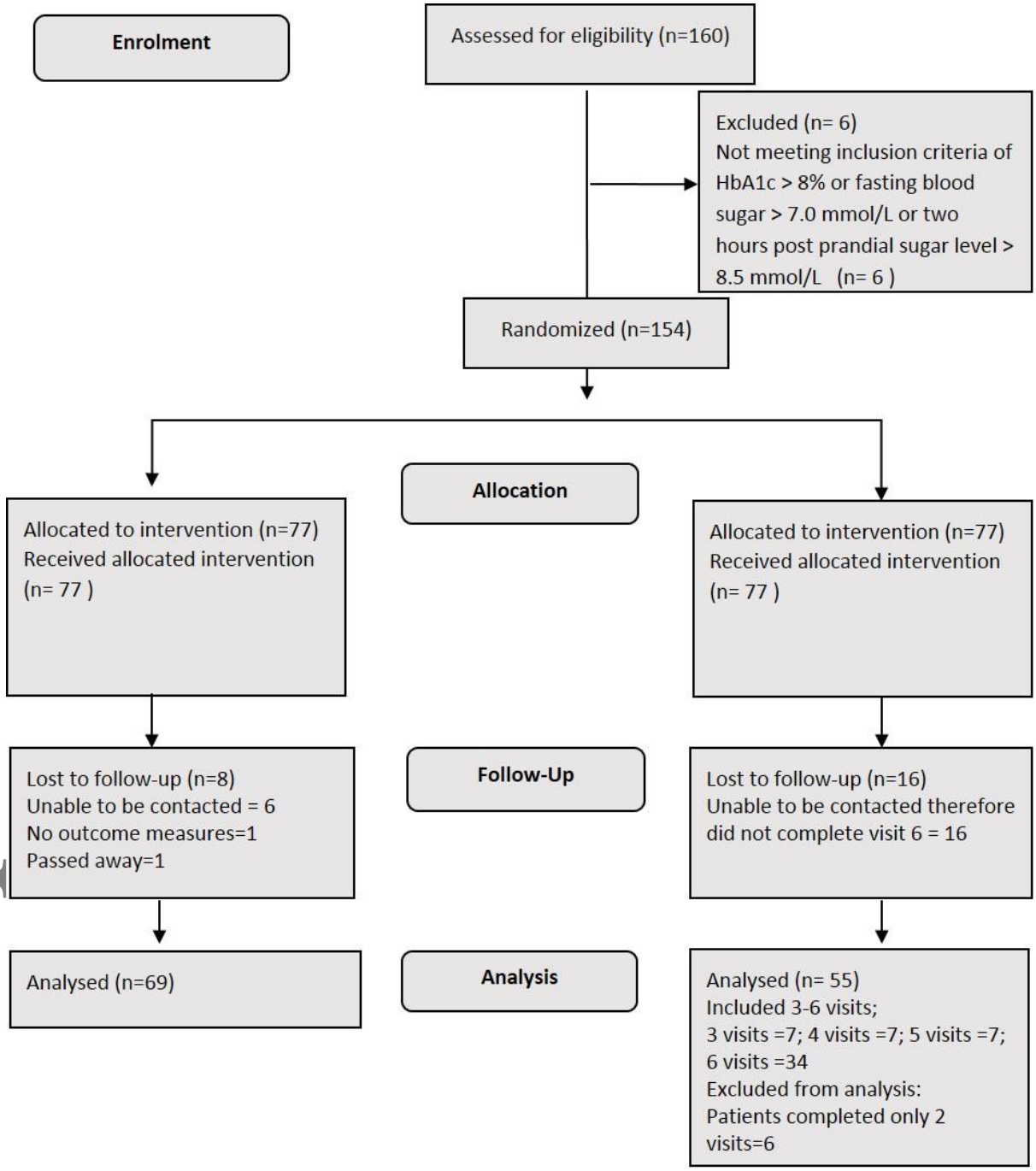

Figure 2: CONSORT flow diagram of patient recruitment and completion

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Table 1: The Simpler™ pharmacist diabetes intervention tool

S=Statin	• Statin **initiation in patients with CVD** • **†Achieve targets for LDL and TG** • **Statin initiation in patients** > 40 years **old without CVD**
I=Insulin/Glycaemic control	• **Insulin** initiation if glycaemic control not achieved despite being on two or more oral hypoglycaemic agents • Target of **HbA1c ≤ 7%** if no other complications • Management of **hypoglycaemia** • **‡Self-monitoring** of blood glucose • Aim a reduction of **HbA1c by 1%** if above target HbA1c • Initiate/continue **metformin** if not contraindicated
M=Medication	• Assess **medicine related problems** • Review medication adherence using ¶medication compliance questionnaire
P=Blood Pressure	• §Achieve BP target • **ACEI/ARB** initiation in patients with/without microalbuminuria /proteinuria • **Reduce sodium** intake (<2400mg sodium/day; 6g/1 teaspoon/day) • One or more antihypertensive medicine to be taken at **bedtime**
L=Lifestyle	• **Exercise:** 30 mins walking (or equivalent) 5 or more days/week (total ≥150 min/week) • **Weight loss**: Caucasian (BMI< 25 kg/m2), Asian (BMI ≤ 23 kg/m2) • **Smoking cessation** • **Waist circumference**: Caucasian (<94 cm in men,<80 cm in women, Asian (≤90 cm in men,≤80cm in women) • **Alcohol** intake: ≤2 standard drinks (20 g) per day for men • Management of **stress**& diabetes related distress • **Erectile dysfunction**: recommend Phosphodiesterase-5 inhibitor as first line therapy for male patients • **Foot care** • Diet advice using **plate model** • Annual **eye** assessment • Address **sleep** hygiene
E=Education	• Knowledge &**understanding** of medicine • Medicine **storage** • Medication optimisation during **fasting month** for Muslims and other religious groups
R=Cardiovascular Risk	• **Aspirin** therapy as secondary prevention in those with diabetes with history of CVD • Use of **Framingham risk calculator** to calculate CVD risk and educate patients • ††Aspirin therapy (75mg-162mg/day) as primary prevention to decrease CVD risk (10 year risk>10%, Framingham)

†Australia: Low density lipoprotein(LDL) <2.0 mmol/L, Triglycride (TG) <2.0 mmol/L, Malaysia: LDL <2.6 mmol/L, TG <1.7 mmol/L

‡Australia:(6.0-8.0 mmol/L fasting),(8.0-10.0 mmol/L-2h postprandial); Malaysia:(4.4-7.0 mmol/L fasting),(4.4-8.5 mmol/L-2h postprandial)

§Australia:≤140/90 mmHg, with albuminuria/proteinuria<130/80 mmHg; Malaysia: ≤135/75 mmHg

¶Medication complicance questionnaire (MCQ) consists of seven questions and has been validated for the Malaysian population.[1]

††Recommendations according to 2016 ADA Standards of medical care in diabetes[2]; Malaysia diabetes clinical practice guidelines recommend aspirin therapy if 10 year risk>10% only for patients aged 65 years and above[3]
ACEI=Angiotensin converting enzyme inhibitors; ARB= Angiotensin 11 receptor blockers; BP= Blood pressure; BMI=Body mass index; CVD=Cardiovascular disease; HbA1c=glycosylated haemoglobin and reflects average glycaemia the preceeding 6-8 weeks LDL=Low density lipoprotein; TG=Triglyceride

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Table 2: Baseline characteristic of the study population

Characteristics	Simpler™ care n (%)	Usual care n (%)	*p*-value
Mean age(years)	55 (SD=8.9)	58 (SD=10)	0.068
Existing family history of diabetes	29 (53.7)	21 (38.2)	0.106
Gender			**0.854**
Female	32 (58.2)	39 (56.5)	
Male	23 (41.8)	30 (43.5)	
Ethnic origin			**0.837**
†*Melayu*	43 (78.2)	53(76.8)	
Chinese	8 (14.5)	11 (15.9)	
Indian	4 (7.3)	5 (7.2)	
Comorbidities			
Hypertension	49(76.6)	61 (69.3)	0.905
CVD	1 (1.6)	1 (1.1)	
Nephropathy	5 (7.8)	12 (13.6	0.182
Retinopathy	3 (4.7)	3 (3.4	
Neuropathy	3 (4.7)	1 (1.1)	
≥2 comorbidities	12 (21.8)	21 (30.4)	0.393
Highest education level			***0.028**
Primary	10 (18.2	29 (42.0)	
Secondary	34 (61.8)	34 (49.3)	
Diploma	6 (10.9)	3 (4.3)	
University	3 (5.5)	1 (1.4)	
Not educated	2 (3.6)	2 (2.9)	
Current employment			**0.080**
Caring for family	18 (32.7)	34 (49.3)	
Working	28 (50.9))	30 (43.5)	
Unemployed	2(3.6%)	0 (0%)	
Retired	7 (12.7%)	5 (7.2%)	
Clinical parameter	**SC mean (SD)**	**UC mean (SD)**	**p-value**
HbA1c (%)	10.68 (2.2)	10.32 (1.5)	0.337
Systolic BP (mm Hg)	136.97(16.8)	137.82 (18.3)	0.791
Diastolic BP(mm Hg)	80.95(12.0)	79.87 (10.5)	0.592
LDL (mmol/L)	3.09(1.1)	3.35 (1.2)	0.317
HDL (mmol/L)	1.28(0.3)	1.19 (0.2)	0.158
TG (mmol/L)	1.93(1.2)	1.97 (0.9)	0.870
Creatinine(mmol/L)	78.46(24.2)	80.74 (25.1)	0.657
BMI (Kg/m^2)	28.82 (5.43)	29.32 (5.3)	0.606
Waist circumference (cm)	96.51(11.3)	95.78 (10.7)	0.717

†*Melayu* refers to ethnic Malay and indigenous population in Malaysia

*Chi-square and Fisher's exact tests was used to determine significance, denoted as $p<0.05$, between intervention (SC) and usual care (UC) arms.

BP=blood pressure; BMI=Body mass index; CVD=cardiovascular disease; HbA1c= Hemoglobin A1c; LDL=Low density lipoprotein; HDL=High density lipoprotein; TG=Triglyceride; SD=Standard deviation

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Table 3: Clinical parameters of participants at baseline and on 6 months follow up in SC and UC

Variables	SC vs UC	†*n*	Baseline Mean (SD)	6 months Mean (SD)	Mean difference (95% CI)	Baseline vs 6- months **p*-value	SC vs UC (6 months) ***p*-value
HbA1c (%)	SC	49	10.68 (2.22)	9.08 (2.26)	-1.59 (-2.20 to -0.90)	***<0.001	***<0.001
	UC	63	10.32 (1.50)	10.07 (1.80)	-0.25 (-0.62 to 0.11)	0.171	
BMI (kg/m$^{2)}$	SC	55	28.82 (5.43)	28.72 (5.33)	-0.1 (-0.60 to 0.40)	0.710	0.481
	UC	68	29.32 (5.26)	29.52 (5.22)	0.20 (-0.40 to 0.80)	0.529	
Waist circumference (cm)	SC	54	96.51(11.31)	95.02(8.84)	-1.49(-3.60 to 0.60)	0.164	0.249
	UC	65	95.78 (10.71)	95.71 (11.17)	-0.06 (-1.44 to 1.31)	0.928	
Systolic BP (mmHg)	SC	55	136.97(16.78)	130.68(14.35)	-6.28(-10.50 to -2.00)	***0.005	***0.028
	UC	69	137.82 (18.33)	138.07(13.35)	0.26 (-3.74 to 0.43)	0.897	
Diastolic BP (mm Hg)	SC	55	80.95(12.00)	78.49(9.65)	-2.46(-5.80 to -0.80)	0.141	0.705
	UC	69	79.87 (10.47)	78.21 (9.15)	-1.66 (-4.37 to 1.05)	0.225	
Total Cholesterol (mmol/ml)	SC	44	5.19(1.25)	5.02 (1.19)	-0.17(-0.60 to 0.30)	0.419	0.943
	UC	53	5.36 (1.30)	5.21(1.43)	-0.15 (-0.50 to 0.20)	0.373	
TG (mmol/ml)	SC	44	1.93(1.23)	1.67(0.94)	-0.26(-0.60 to 0.10)	0.138	0.202
	UC	54	1.97 (0.92)	1.99 (1.15)	0.03 (-0.26 to 0.31)	0.856	
LDL (mmol/ml)	SC	38	3.09(1.13)	2.61(1.09)	-0.49(-0.90 to -0.10)	***0.017	0.553
	UC	49	3.35 (1.16)	3.01 (1.25)	-0.33(-0.68 to 0.01)	0.058	
HDL (mmol/ml)	SC	37	1.28(0.34)	1.40(0.39)	0.12 (1.20 to 2.30)	***0.010	0.598
	UC	50	1.19 (0.24)	1.28 (0.30)	0.09 (0.01 to 0.17)	***0.024	

†Number of patients with an available result

**p* values from paired t-test. Changes over six months are mean difference (95% CI)

***p* values from independent t-test.

***statistically significant

SC=Simpler™ care; UC= Usual care; SD=standard deviation; CI= confidence interval; HbA1c= haemoglobin A1c; BMI=body mass index; BP=blood pressure; TG=triglyceride; LDL=low density lipoprotein; HDL=high density lipoprotein

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Table 4: Comparison of participants in SC and UC arms who met treatment target at 6 months according to the 2015 Malaysian clinical practice guidelines on management of T2DM.

Variables	Arm	n	Total patients (%)	*p-value SC vs UC
HbA1c ≤ 6.5%/48mmol/mol	SC	49	7(14.3)	**0.020
	UC	65	1(1.5)	
Systolic BP target: ≤135 mmHg	SC	55	44 (80)	**0.001
	UC	69	29 (42)	
Diastolic BP target: ≤75 mmHg	SC	55	15 (27.3)	0.577
	UC	69	22 (31.9)	
TG ≤ 1.7 mmol/L	SC	45	32 (71.1)	0.080
	UC	59	32 (54.2)	
LDL ≤ 2.6 mmol/L	SC	43	26 (60.5)	**0.046
	UC	57	23 (40.4)	
HDL > 1.0 (male)	SC	17	13 (76.5)	1.000
	UC	22	17 (77.3)	
HDL > 1.2 (female)	SC	20	15 (75.0)	0.226
	UC	28	15 (53.6)	

*p-value from Pearson chi-square test; **statistically significant; SC=Simpler™ care; UC=usual care; SD=standard deviation; CI=confidence interval; CVD=cardiovascular disease; HbA1c= haemoglobin A1c; BMI=body mass index; BP=blood pressure; TG=triglyceride; LDL=low density lipoprotein; HDL=high density lipoprotein

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Table 5: Comparison of patients who achieved at least a 1% decrease in HbA1c

Variable	No. patients (%)	*p-value
Overall		
SC (n=49)	28 (57.1%)	**0.002
UC (n=63)	18 (28.6%)	
Patients of *Melayu* ethnicity		
SC (n=49)	23 (46.9%)	**0.003
UC (n= 63)	13 (20.6%)	
Patients of non *Melayu* ethnicity		
SC (n=49)	5 (10.2%)	0.349
UC (n=63)	5 (7.9%)	
Female patients		
SC (n=49)	16 (32.7%)	**0.006
UC (n=63)	9 (14.3%)	
Male patients		
SC (n=49)	12 (24.5%)	0.136
UC (n=63)	9 (14.3%)	
Patients with mean baseline HbA1c≤9.0%		
SC (n=49)	5 (10.2%)	0.204
UC (n=63)	3 (4.8 %)	
Patients with mean baseline HbA1c>9.0%		
SC (n=49)	23 (46.9%)	**0.008
UC (n=63)	15 (23.8%)	

*p-value from Pearson chi-square test; **statistically significant; *Melayu* refers to ethnic Malay and indigenous population in Malaysia
HbA1c= glycated haemoglobin;

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